

ISTAT traffic accidents analysis

Goal:

Find the italian cities with the most accidents.

Fetch Data:

ISTAT API

- City id
- Year
- Accidents data



RAW DATA

SITUAS .csv

- City name
- Province
- Region name
- Residents
- Area

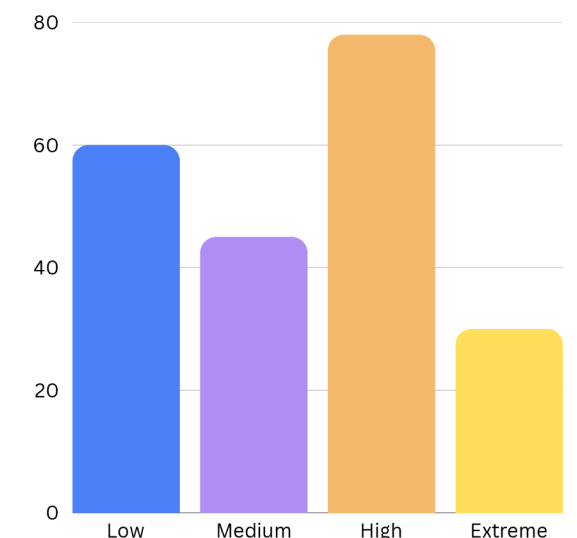


Combine the information from both data sets based on the ID code of each city.

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Data Preparation:

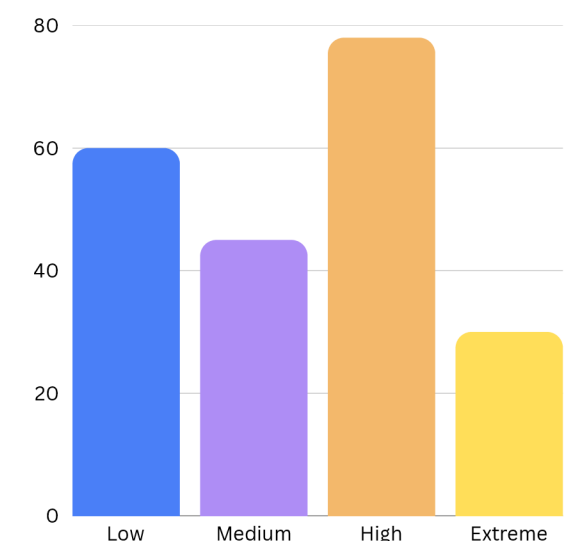
- Check the data and fix missing values and *Nan* values.
- The data was prepared for analysis by removing and renaming columns and fixing data types to improve memory performance.
- Save the fixed and cleaned dataset as a .csv file.



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Analysis:

- Change the data frame's structure to make it easier to analyze and format the column names.
- Calculate the metrics (*per capita accidents*, *severity index*, *fatality rate* and *collision density*).
- Look at how things have changed over time.
 - 2020 low pick course, caused by the pandemic.
- Select Years 2021-2024 and create a risk score by combining metrics into a single variable to help identify the most dangerous cities.



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Hypothesis Validation:

- Check the top 20 cities to identify patterns and make initial assumptions.
- Perform a permutation test to validate or refute those hypotheses.
- Report the conclusions of the analysis.
- Incorporating graphs into a Power BI report.

